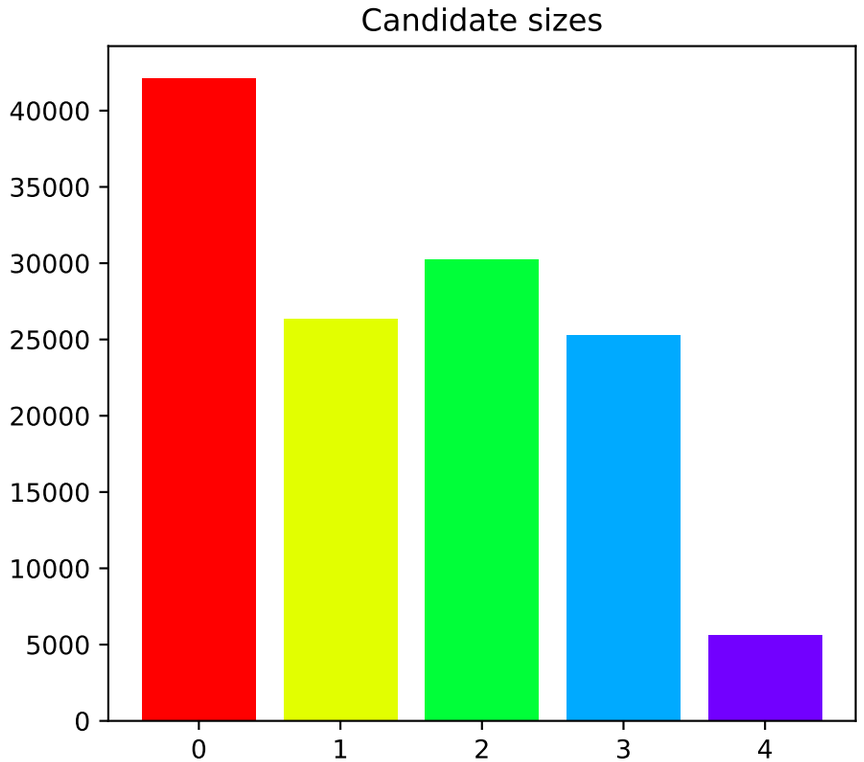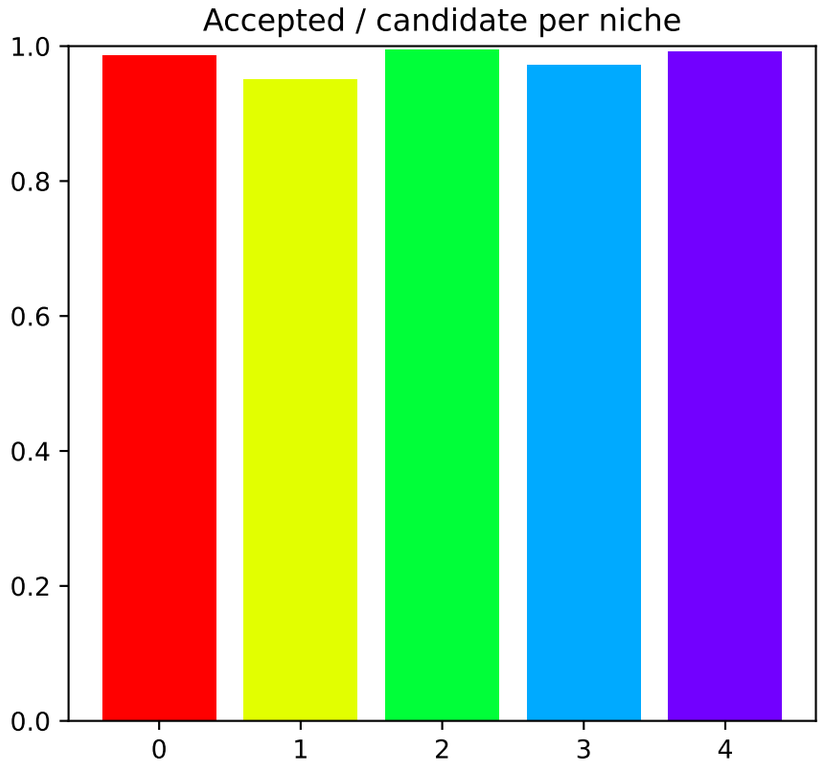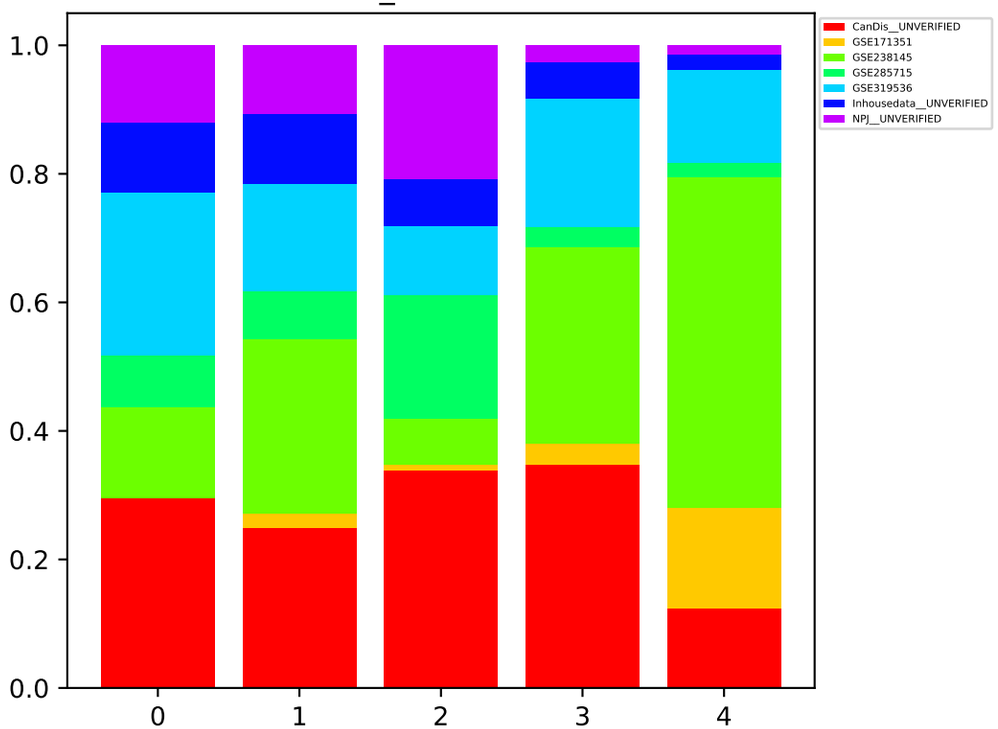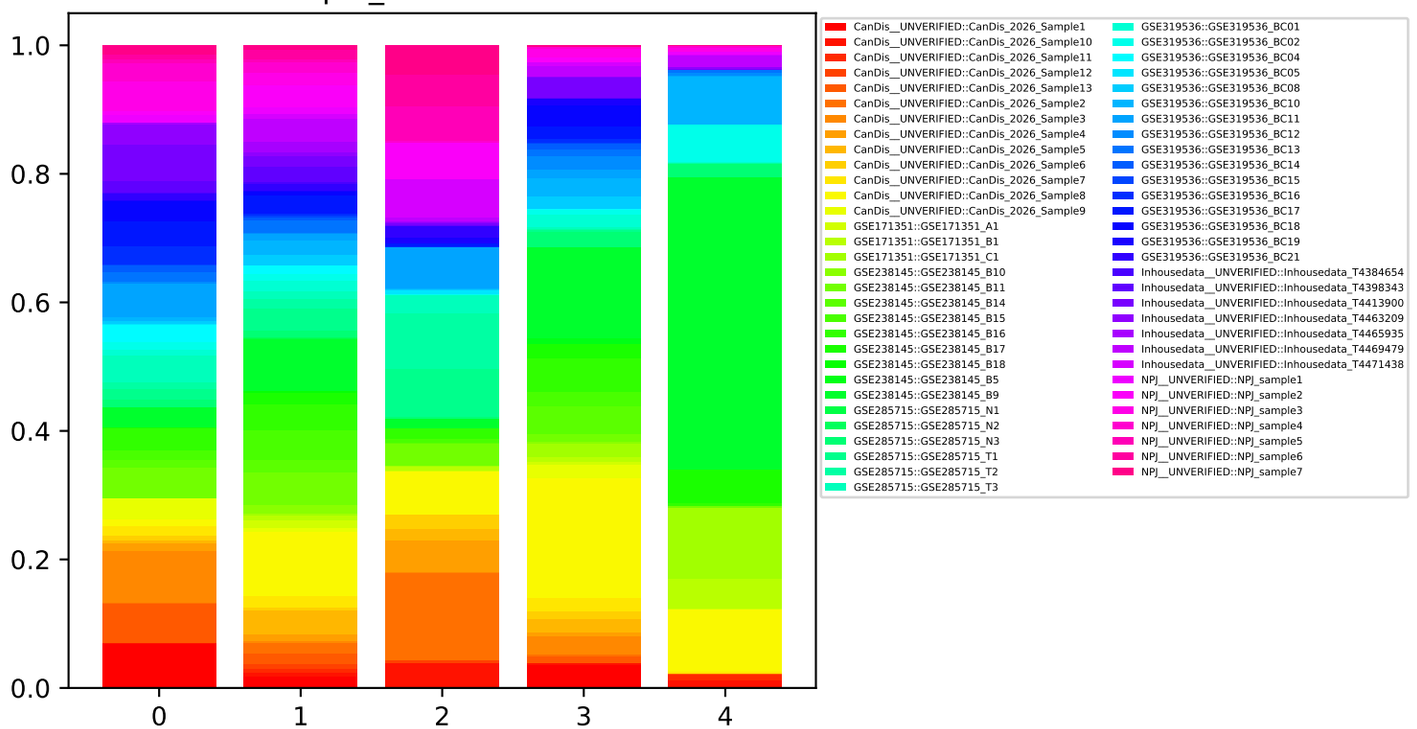SAMPLE_R2_LEIDEN